

Abstract

AC to DC power converters are devices that allow stable power for electronic devices to function and are extremely integral for safe usage of everyday devices. In this work, we have fabricated an AC to DC power converter with 5 stages that create a DC voltage power supply. The first stage is the astable multivibrator, which consists of a non-inverting op-amp, a capacitor with a value of $4.7\mu\text{F}$, and two resistors in parallel with values of $1\text{ k}\Omega$ and $5.75\text{ k}\Omega$, respectively. These values were chosen to ensure that the triangle waves were created, as the Astable Multivibrators are circuits that do not have stable states since the capacitors are continuously charging and discharging. The stimulated peak to peak (P.P.) value for the Astable multivibrator was 26V , while the lab P.P. value was 27.3V . Note that the op-amp built in this experiment does not have negative feedback. The astable multivibrator is connected to a Full Wave Bridge Rectifier that contains four diodes, two of which are in forward bias, and the other two which are in reverse bias, assuming a positive clock cycle. A load resistor was attached in parallel across the full-wave bridge rectifier to measure the raw DC voltage. The stimulated peak to peak (P.P.) value for the rectifier was 6.4 V , and the experiment value was 8V . Additionally, the last three stages that are cascaded together and which complete the final build of the power converter circuit are the RC low pass filter, the zener diode voltage regulator, and the load resistor. The lowpass RC circuit filter, built with a resistor value labeled $1\text{ k}\Omega$ and a capacitor value of $4.70\text{ }\mu\text{F}$, serves as a device to filter the voltage ripples that are retrieved from the full wave bridge rectifier. It's P.P. theoretical value was roughly 4V , while the experimental value was 4.8 V . The zener diode's function was to maintain a constant DC voltage supply with an experimental peak to peak

value of 280 mV and a theoretical value of 360 mV. The load resistor's function was just to measure the output voltage from the zener diode regulator. The entire circuit was tested in the Electronics Lab at the University of Houston.

Listed below are the calculations and percent errors for the peak to peak voltages for all four stages :

$$\% \text{ error} = ([\text{Accepted Value} - \text{Observed Value}] / \text{Observed Value}) * 100;$$

Astage Multivibrator

$$\% \text{ error} = (27.3[\text{V}] - 26[\text{V}]) / 26 * 100 = 5\%$$

Diode Rectifier

$$\% \text{ error} = (6.4[\text{V}] - 8[\text{V}]) / 8 * 100 = 20\%$$

Filter

$$\% \text{ error} = (4 [\text{V}] - 4.8[\text{V}]) / 4.8 * 100 = 17\%$$

Voltage Regulator

$$\% \text{ error} = (3.6[\text{V}] - 2.8[\text{V}]) / 2.8 * 100 = 29\%$$

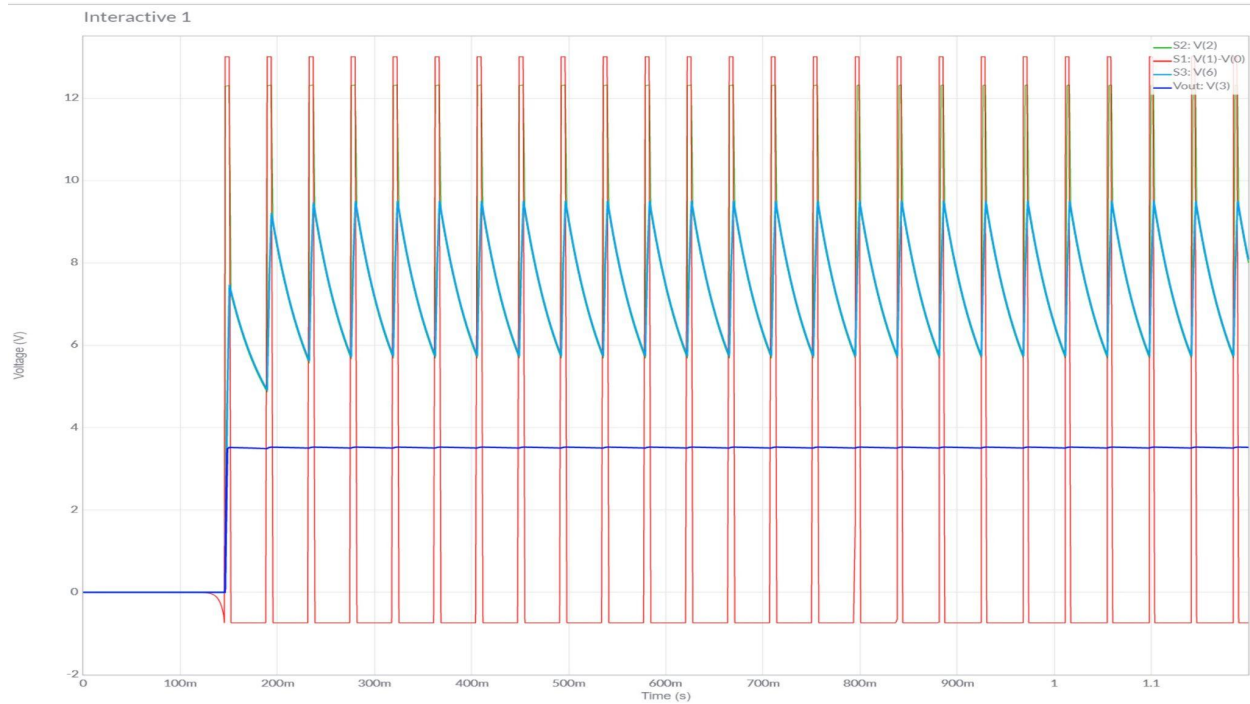
INTRODUCTION

Power sent from electric power grid stations to commercial, industrial, and residential buildings is usually inconsistent since power usage varies throughout different real estate depending on demand. A solution to this problem is a AC to DC power converter that reduces the ripples created from the sinusoidal waves. An AC to DC converter can be made from 5 modules in the lab as listed : For the astable multivibrator, the specifications of this project include a UA741 non-inverting op-amp and 3 resistors which range between 1k Ω to 6 k Ω . For the diode bridge rectifier, 4 1N4001 Diodes were used. For the RC circuit filter, one 1k Ω resistor and one 4.7 μ F capacitor was used. The zener diode module, 1N5227B, was connected in parallel with the load resistor attached to the output of the circuit.

Theoretical Considerations

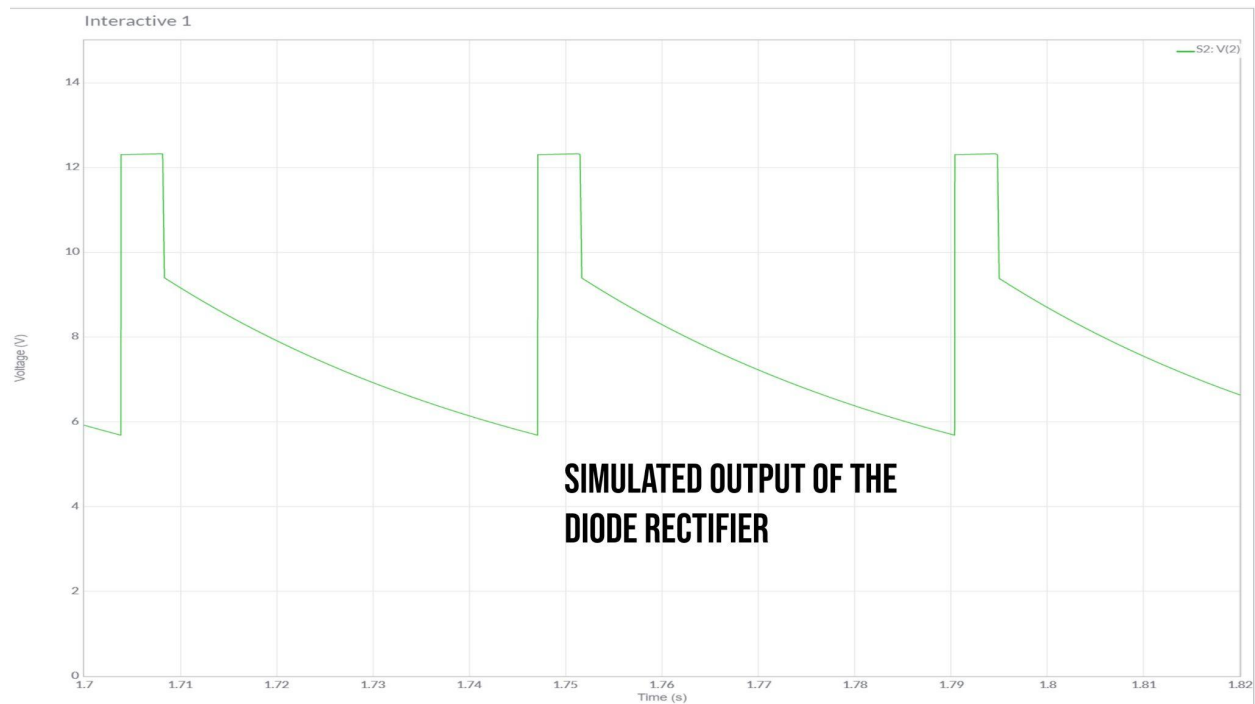
- a. Attached below is the circuit diagram for the device, including the specifications :

In this work, the astable multivibrator does not have a non-inverting op-amp. A square waveform is generated, and an indication that the waveform changes its state is when the output voltage, which is connected to the capacitors that charge and discharge depending on the voltage, is greater than or equal to the voltage at the non-inverting terminal. In essence, the astable multivibrator serves as an AC voltage input.



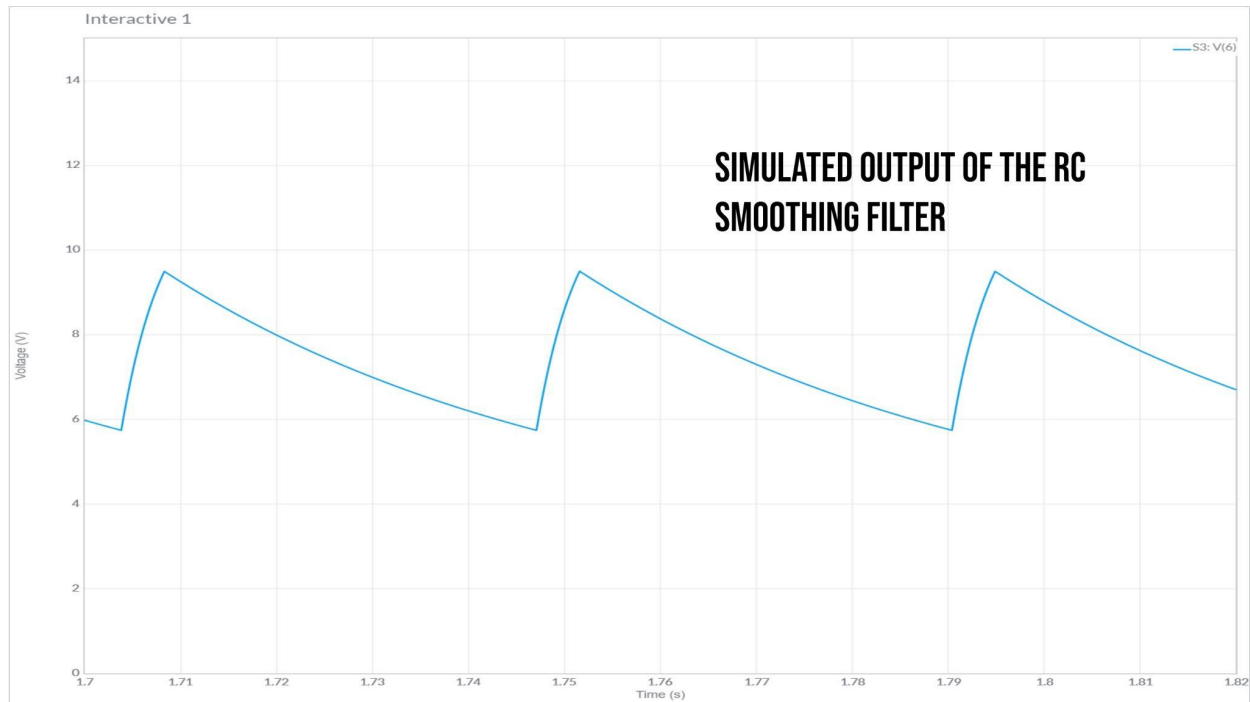
Astable Multivibrator : The voltage rise and fall between time periods indicates that the capacitor is continuously charging and discharging.

The second stage below is the Full Bridge Rectifier. This is an extremely important stage of the circuit since the primary transformation of an AC signal to an unfinalized DC signal occurs. There are 4 diodes, labeled D1, D2, D3, and D4. Assuming a positive cycle, D1 and D2 are in forward bias and are the only diodes to conduct current. Contrastingly, D3 and D4 are in reverse bias and do not conduct current. It's important to note that there is a maximum peak value of where the square waveform reaches.



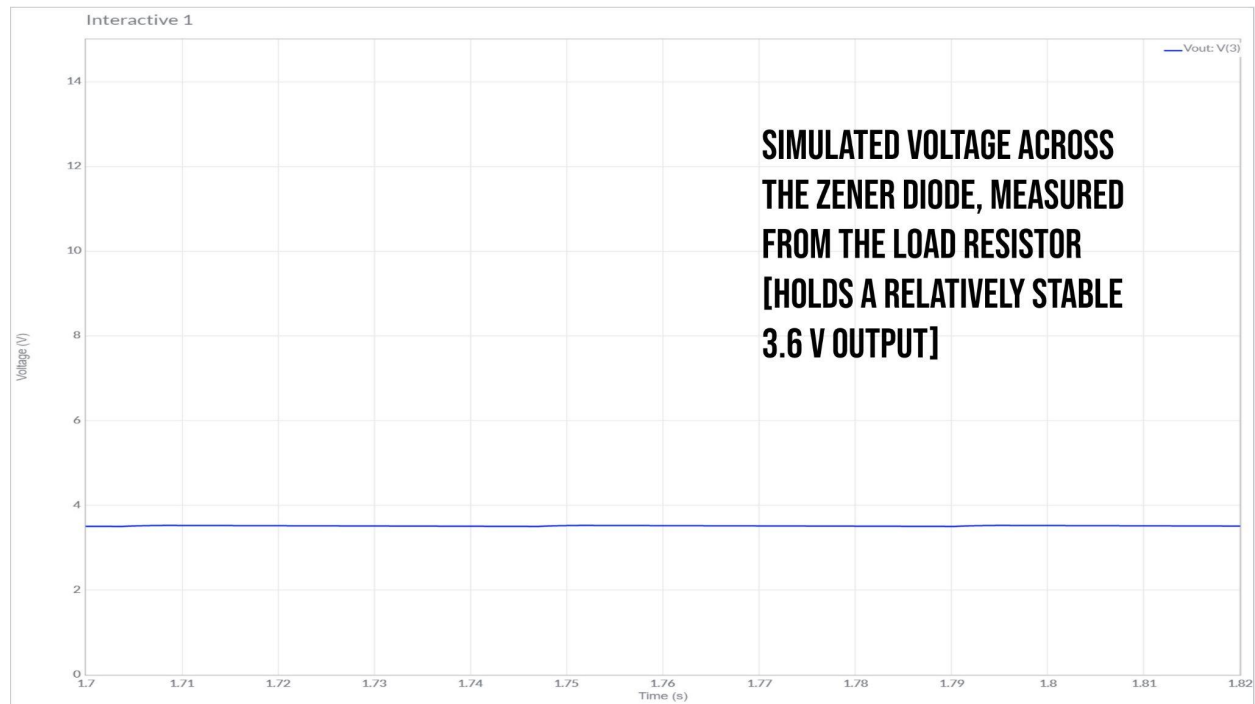
Diode Rectifier : The output voltage is starting to stabilize through the diode rectifier.

The third stage is the RC Circuit Filter, more specifically, a low pass filter type. This stage is equally as integral as stage 2 as the filter begins the process of smoothing out the ripples from the raw DC voltage. The low pass filter was chosen to convert out the higher, but inconsistent voltage values, and stabilize the voltage.



RC Low Pass Smoothing Filter : The output voltage is starting to further stabilize through the RC Low Pass filter, and the ripples are starting to flatten out.

The fourth stage of this circuit is the Zener Diode Regulator, which is used to keep the DC voltage consistent with minimal to no alterations. Zener diodes function in reverse bias, and when it reaches the breakdown voltage, then the current flowing through the zener diode flows into R_s (source resistor) to prevent the zener diode from overloading and to keep a stable DC voltage output. Hence, a constant DC voltage supply is maintained, and the zener diode serves as a type of voltage source.



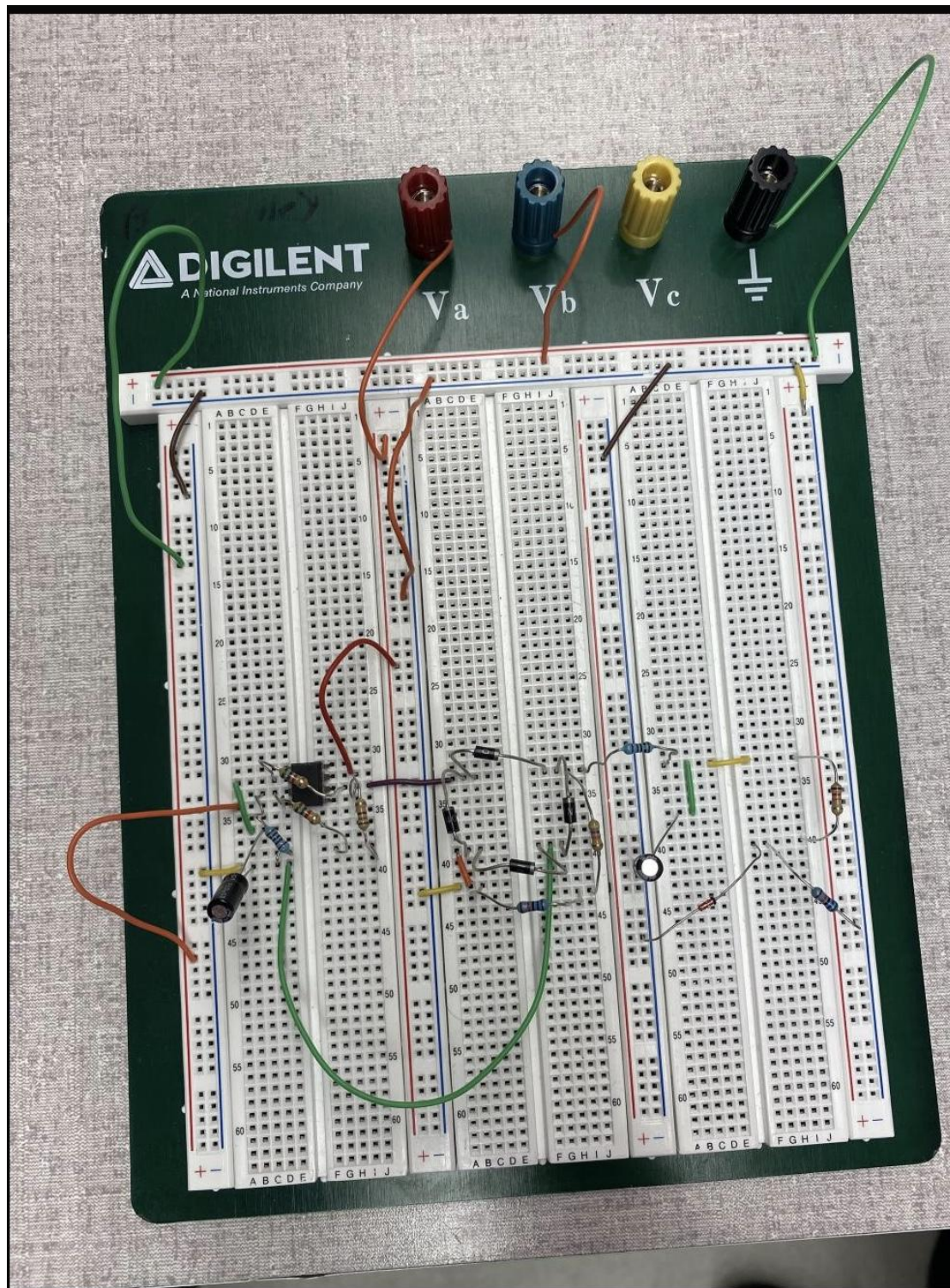
Zener Diode : The output voltage has stabilized to the maximum with little to no ripples.

Two mini-teams were created to design the AC to DC converter on Multisim and sketch the design on paper, respectively. Once the design on paper was agreed by all of the team members, the Multisim team members designed each of their modules. It's important to note that the modules were split between the members, but all of the modules were completed in a timely manner and were combined to create one giant circuit on April 18th. Next, between April 18th and April 25th, the circuit, which was built at the home of an individual team member, brought the circuit into the Electronics Lab for testing. After numerous debugging, switching resistors and capacitors around, and asking the teaching assistants for guidance regarding the incorrect voltage output for Module 4 and 5, it was later found out that there was a loose wire connected to the circuit board. Additionally, the original zener diodes in the team member's lab kit malfunctioned. By removing the loose wire hanging from the circuit board and

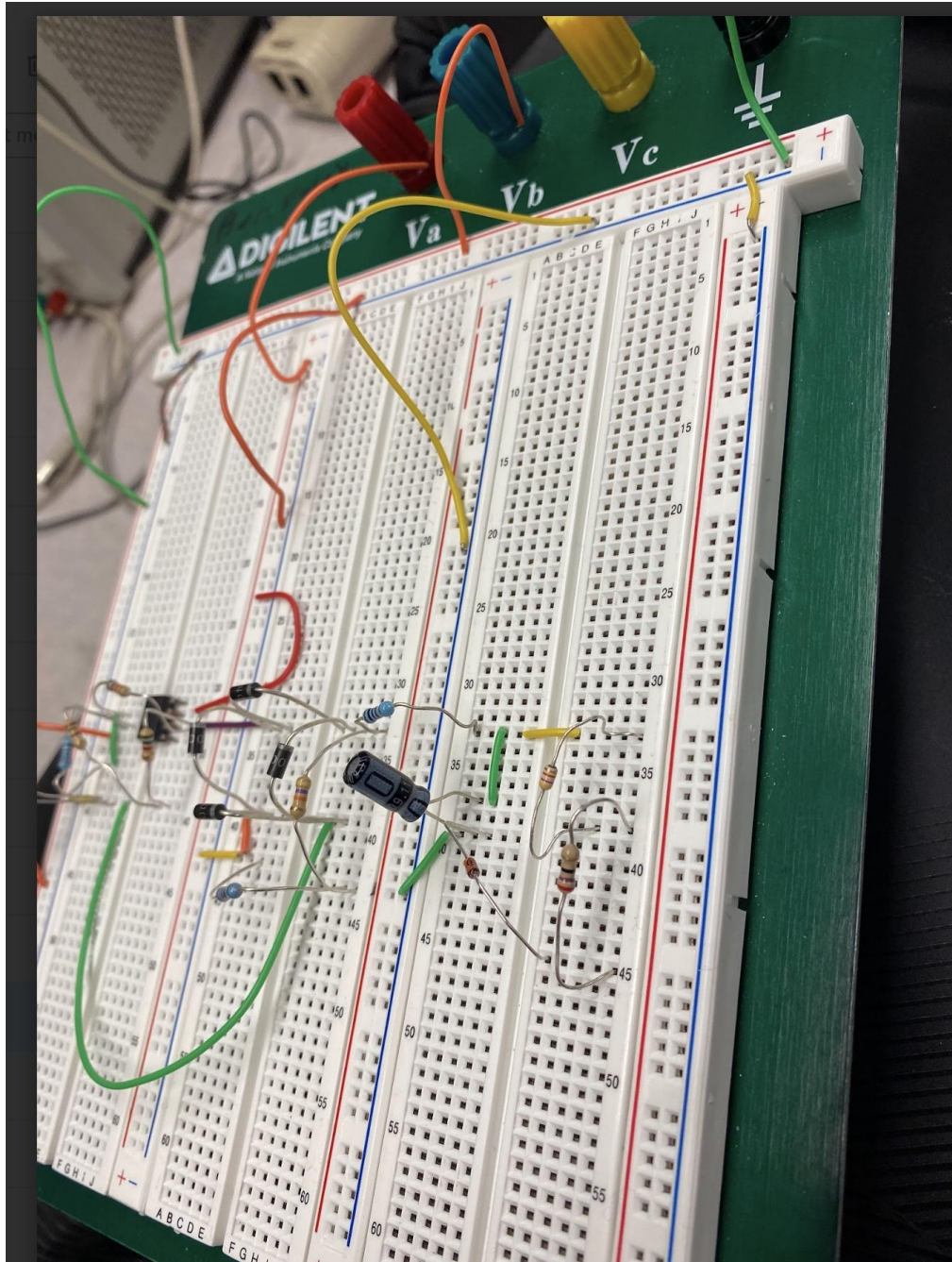
replacing the zener diodes with functional ones in the electronics lab, Stage 4 and Stage 5 matched the results from Multisim.

The AC to DC converter project was due on April 26th. A presentation regarding our project was done by having each individual team member discuss each module and its role in the circuit.

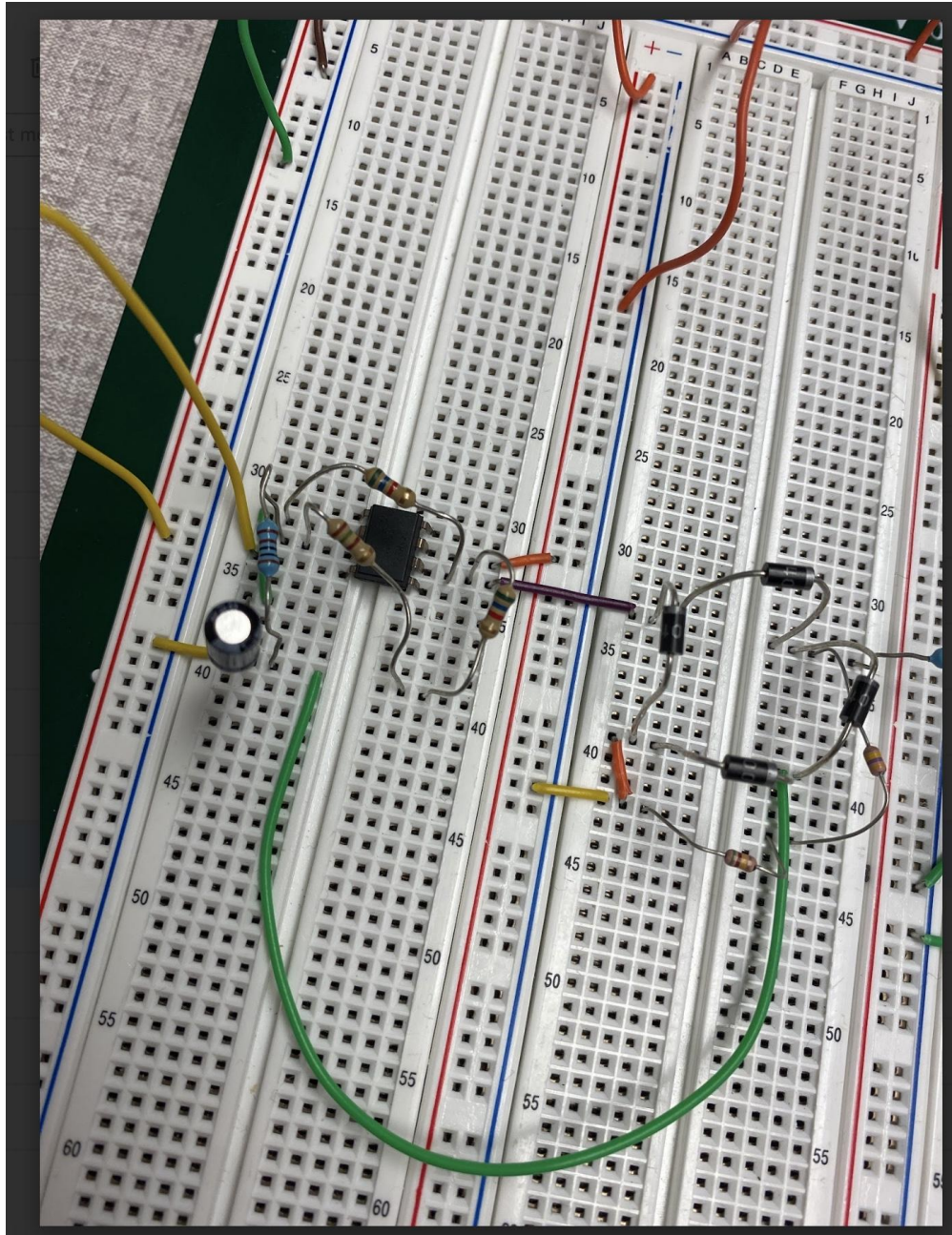
Experimental procedures



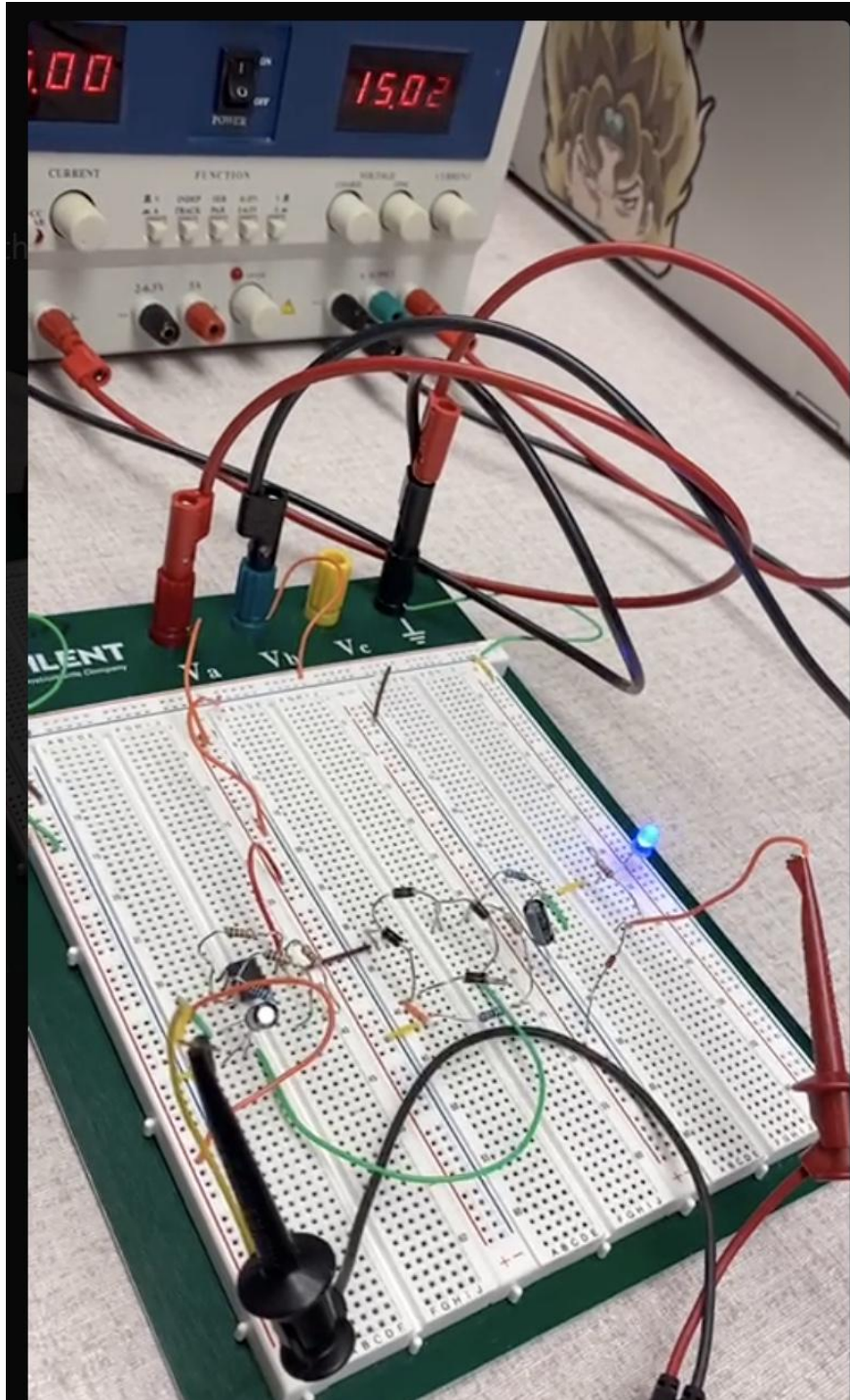
The first top three wires(green, red, and orange) connect the power supply to the breadboard.



There is a 100 k Ω resistor placed in parallel with the rectifier diode output, which is connected to the RC low pass filter. It is a blue resistor placed horizontally. For the RC low pass filter, a 4.7 μ F capacitor and 1k Ω resistor are placed in parallel to further reduce the ripple effects of the voltage output from the rectifier output. Finally, a zener diode is placed in parallel with the 4.7 μ F capacitor and source resistor of 10k Ω . A load resistor located at the far right of the circuit, is placed to measure the final DC output voltage.



Starting from the left is the Astable multivibrator with a non-inverting op-amp with a capacitor entering the negative terminal of the op-amp. The green wire with a quadratic parabola connects the voltage output of the op-amp to the diode bridge rectifier. Four diodes are placed in the form of a square to create a full bridge diode rectifier.

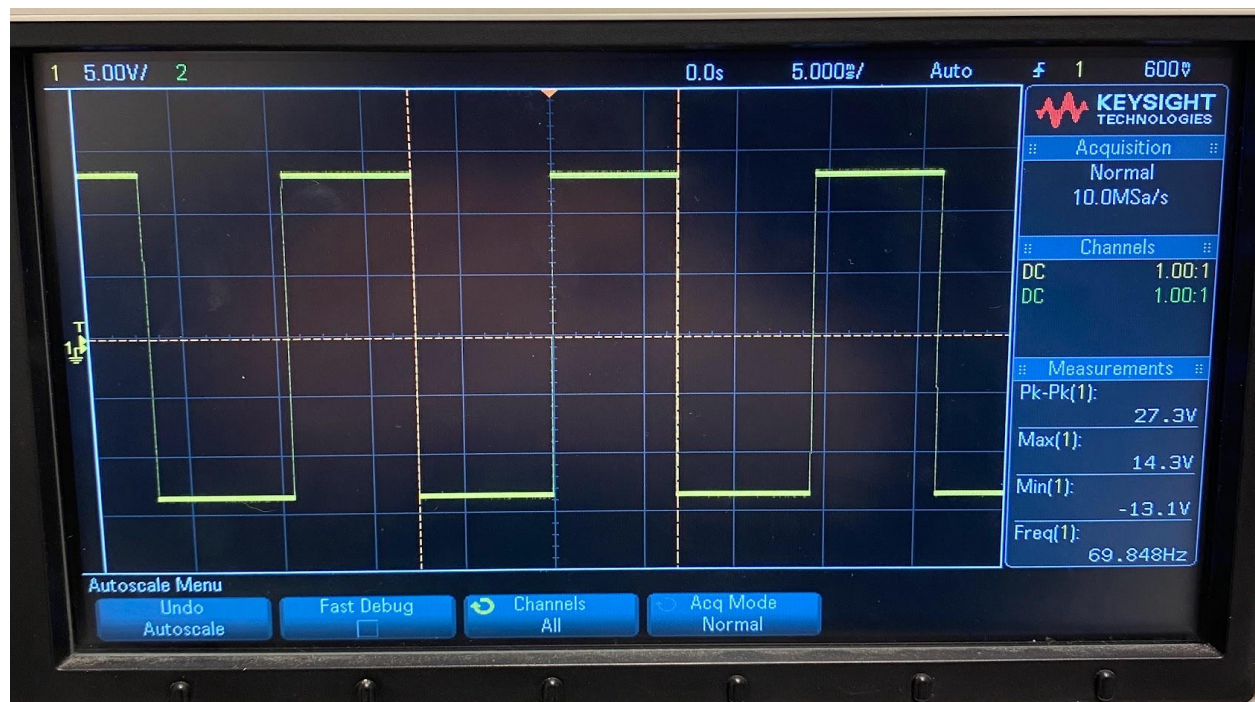


The LED light on indicates that the DC voltage output was functioning, and as a whole, the circuit was functioning correctly.

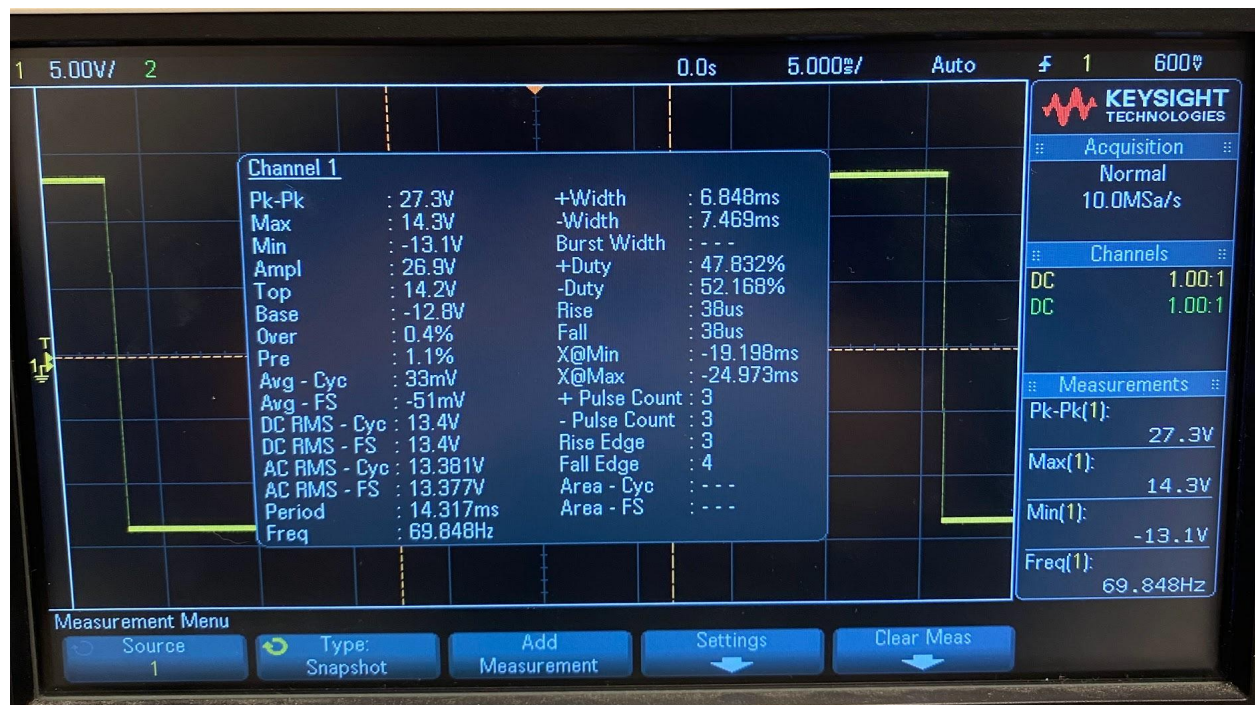
The circuit modules were tested by disconnecting the individual resistors and nodes that connected the output voltages with each module. For instance, to test the astable multivibrator, the green wire in the shape of a quadratic parabola was disconnected from the bridge rectifier and was connected to the power supply and oscilloscope to show the most recent DC voltage waveform. The same applied for the remaining 3 modules.

Results

Below is the test data for the Astable Multivibrator :



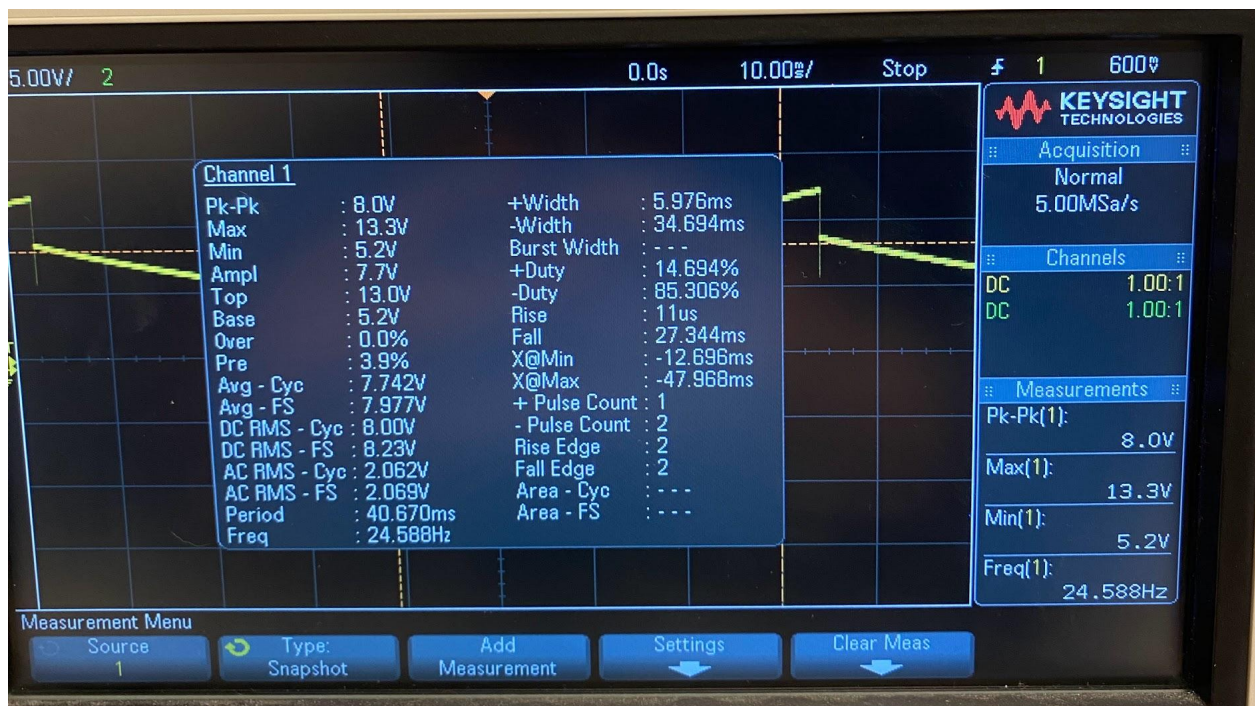
The Peak to Peak value for the astable multivibrator was 27.3[V]. The theoretical peak to peak value, however, was 26[V].



The Peak to Peak value for the astable multivibrator was 27.3[V]. The theoretical peak to peak value, however, was 26[V](graph specifications).



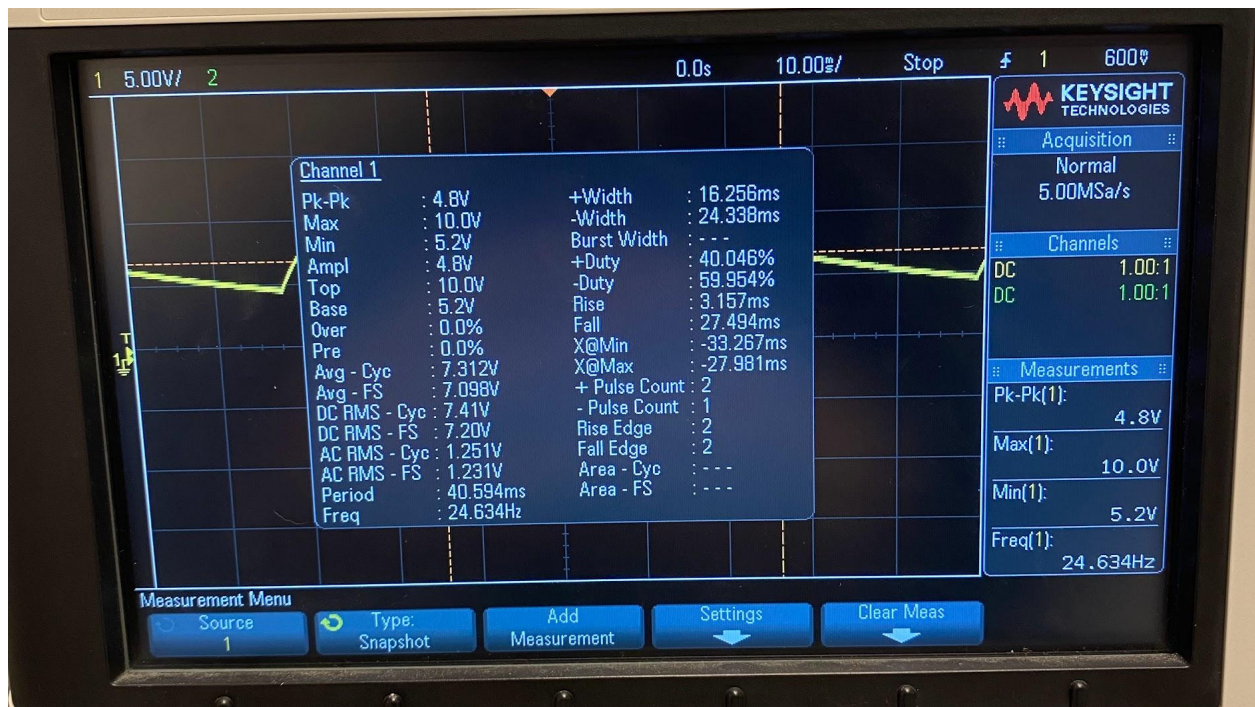
The Peak to Peak value for the full bridge rectifier was 8[V]. The theoretical peak to peak value, however, was 6.4[V]



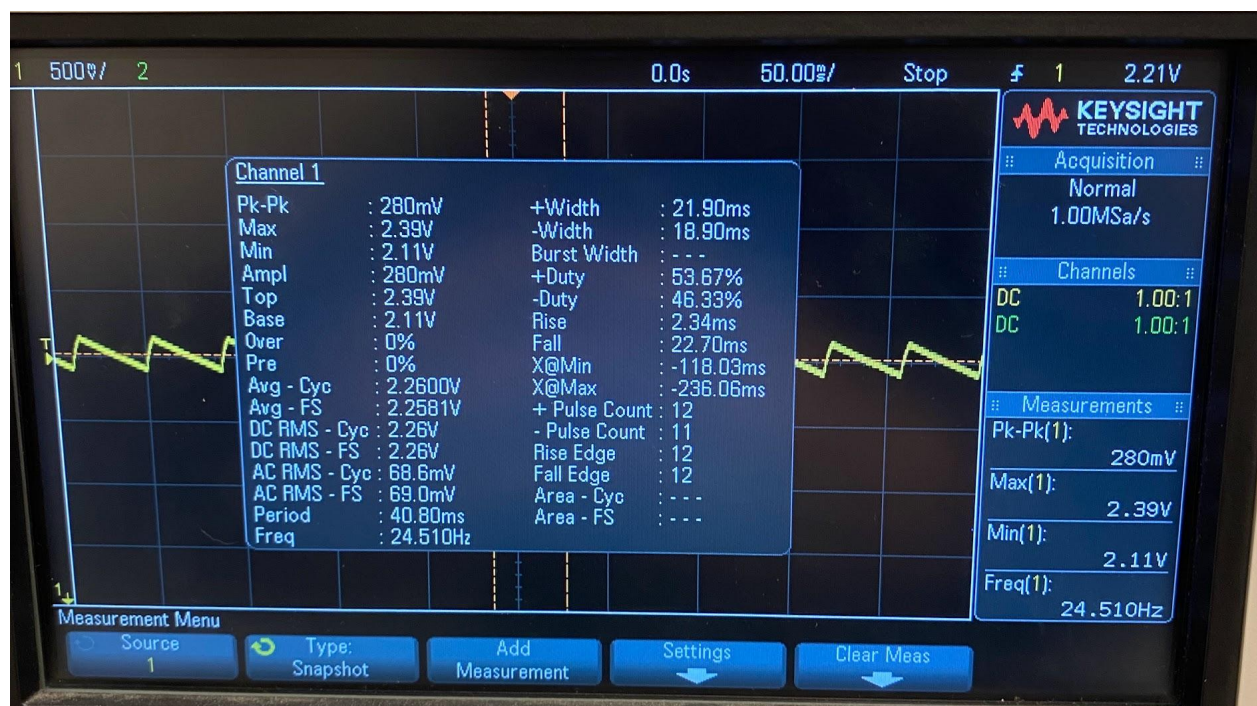
The Peak to Peak value for the full bridge rectifier was 8[V]. The theoretical peak to peak value, however, was 6.4[V](graph specifications).



The Peak to Peak value for the astable multivibrator was 4.8[V]. The theoretical peak to peak value, however, was 4.0[V].



The Peak to Peak value for the astable multivibrator was 4.8[V]. The theoretical peak to peak value, however, was 4.0[V].



The Peak to Peak value for the astable multivibrator was 2.8[V]. The theoretical peak to peak value, however, was 3.6[V].



The Peak to Peak value for the astable multivibrator was 2.8[V]. The theoretical peak to peak value, however, was 3.6[V].

Discussions

The AC to DC converter project took 1 month to complete. The 2 major milestones of this project were the construction of the AC to DC converter on Multisim, and testing the built circuit in the Electronics lab. The multisim circuit

design was completed on April 18th. However, the physical build of our circuit was re-done numerous times as we found potential faults in our build. Such cases were that there were 2 malfunctioned diodes used in our experiment only to find out that diode replacements were needed for the Full Bridge Rectifier. Additionally, a loose wire was hanging off the breadboard, which was causing Stage 3, Stage 4, and Stage 5 to depict incorrect DC voltage graphs, different from what was depicted on the Multisim graphs. The team's response to the major issues was spending time in the Electronics Lab debugging the circuit using different diodes, resistor values, and capacitor values. Additionally, 2 zoom call meetings were scheduled with Dr. John Wolfe regarding a deviation of the ripple effects. Although it deviates slightly from the original plan and timeline, the AC to DC converter does meet the high level specifications mentioned in the introduction.

The data collected from Multisim versus the data collected from the Electronics Lab did show discrepancies measured in percent error(please refer to the Abstract for the percent errors). All the modules were functioning correctly.

SUMMARY/ CONCLUSION

My role in this project was to set up a Multisim group such that members working on their specific modules could share their modules when completed. Additionally, I assisted in combining the completed modules in Multisim from individual team members as well as creating a Microsoft Excel graph to depict the relationship between Ripple Amplitude vs. Load Resistance. Finally, in the

Electronics Lab, I helped out with choosing realistic resistor values and capacitor values that were available in the lab such that the modules depicted similar graphs to what was shown on Multisim. At the end of the project, I was requested to discuss the role of the Load Resistor as part of the project presentation. We conclude, therefore, that the integration of the Load Resistor in parallel with a zener diode is an extremely important component to properly depict the accurate DC output voltage of the entire circuit as well as to ensure that the zener diodes are functioning correctly as what was simulated in Multisim.

REFERENCES

[1] "Power supplies," Electronics Club, <https://electronicsclub.info/powersupplies.htm> (accessed May 8, 2023).

[1] W. Storr, "AC-DC converters including Buck, boost and flyback," Basic Electronics Tutorials, <https://www.electronics-tutorials.ws/connectivity/ac-dc-converters.html> (accessed May 8, 2023).

[3] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*. Fort Worth u.a.: Saunders, 1991.

Acknowledgements

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